

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Unit Test 3

Unit Test 3 - Answer Key

Class: Class 8

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: SQL - Database Fundamentals

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 x 1 = 5 marks)

Q.No	Answer	Explanation
1	b	COUNT() returns the total number of rows
2	c	HAVING clause filters groups after GROUP BY
3	b	DISTINCT returns unique department values only
4	a	ALTER TABLE is used to modify existing table structure
5	a	Option a has correct syntax: SELECT * FROM Students WHERE marks > 80 AND grade = 'A';

Section II: Short Answer Questions (5 x 1 = 5 marks)

1. Write a SQL query to create a table Products.

• Answer:

```
CREATE TABLE Products (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    price DECIMAL(10, 2),  
    quantity INT  
);
```

• Marking: 1 mark for correct CREATE TABLE syntax, 1 mark for all columns with data types.

2. Write SQL queries to insert 3 records and update price.

• Answer:

```
INSERT INTO Products VALUES (1, 'Laptop', 45000.00, 10);  
INSERT INTO Products VALUES (2, 'Mouse', 500.00, 50);  
INSERT INTO Products VALUES (3, 'Keyboard', 1200.00, 30);
```

```
UPDATE Products SET price = 450.00 WHERE id = 2;
```

• Marking: 1 mark for correct INSERT statements, 1 mark for correct UPDATE.

3. What is the difference between WHERE and HAVING?

• Answer: WHERE filters rows before grouping (cannot use aggregate functions). HAVING filters groups after GROUP BY (can use aggregate functions). Example: WHERE salary > 50000 vs HAVING AVG(salary) > 40000.

• Marking: 1 mark for explaining WHERE, 1 mark for explaining HAVING with examples.

4. Write a query to find average price grouped by quantity.

• Answer:

```
SELECT quantity, AVG(price) as avg_price  
FROM Products  
GROUP BY quantity;
```

• Marking: 1 mark for correct GROUP BY, 1 mark for AVG function.

5. Write a query to display top 3 products by price.

- **Answer:**

```
SELECT * FROM Products
ORDER BY price DESC
LIMIT 3;
```

- **Marking:** 1 mark for ORDER BY DESC, 1 mark for LIMIT clause.